

$$1. \quad x[n] = A e^{-\beta n} \cos(\omega_0 n) u[n] = A e^{-\beta n} (e^{j\omega_0 n} + e^{-j\omega_0 n}) / 2 u[n]$$

$$x(e^{j\omega}) = \sum_{n=0}^{\infty} \frac{A}{2} \left(e^{(j\omega_0 - \beta)n} + e^{-(j\omega_0 + \beta)n} \right) e^{-j\omega n} = \frac{A}{2} \left[\frac{1}{1 - e^{(j\omega_0 - \beta - j\omega)}} + \frac{1}{1 - e^{-(j\omega_0 + \beta + j\omega)}} \right]$$

2.

$$(a) \quad y[n] = \sum_k x[k] h[n-k], \quad x[k] = x[n], \quad h[k] = x[-n]$$

$$y[m] = \sum_{n=-\infty}^{\infty} x[n] x[m+n] \Rightarrow y[0] = \sum_n x[n] x[n] = \sum_n |x[n]|^2$$

$$(b) \quad x[n] * h[n] \xrightarrow{\mathcal{F}} X(e^{j\omega}) H(e^{j\omega})$$

$$x[n] \xrightarrow{\mathcal{F}} X(e^{j\omega}) \Rightarrow Y(e^{j\omega}) = X(e^{j\omega}) X(e^{-j\omega})$$

$$x[-n] \xrightarrow{\mathcal{F}} X(e^{-j\omega}) \quad \because x[n] \text{ is real } \therefore X(e^{j\omega}) = X(e^{-j\omega})^*$$

$$Y(e^{j\omega}) = X(e^{j\omega}) X(e^{-j\omega}) = X(e^{j\omega}) X(e^{j\omega})^* \Rightarrow X(e^{-j\omega}) = X(e^{j\omega})^* \\ = |X(e^{j\omega})|^2$$

(c)

$$y[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} Y(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 e^{j\omega n} d\omega$$

Consider

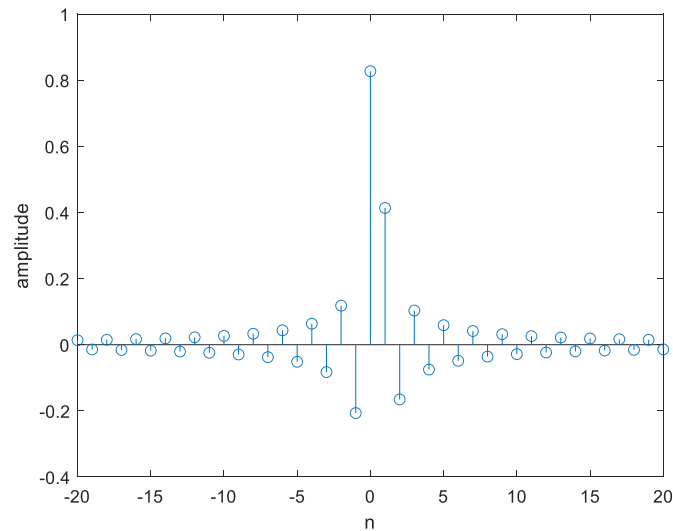
$$y[0] = \sum_n |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

3.

$$(a) \quad H(e^{j\omega}) = e^{-j\omega/3}$$

$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-\frac{j\omega}{3}} e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(n-\frac{1}{3})j\omega} d\omega \\ = \frac{1}{2\pi} \frac{e^{j\pi(n-\frac{1}{3})} - e^{-j\pi(n-\frac{1}{3})}}{j(n-\frac{1}{3})} = \frac{1}{2\pi} \frac{(-1)^n e^{-j\frac{\pi}{3}} - (-1)^n e^{j\frac{\pi}{3}}}{j(n-\frac{1}{3})} \\ = \frac{(-1)^{n+1} (e^{j\frac{\pi}{3}} - e^{-j\frac{\pi}{3}})}{2j\pi(n-\frac{1}{3})} = \frac{(-1)^{n+1} \sin(\frac{\pi}{3})}{\pi(n-\frac{1}{3})} \\ = \frac{\sqrt{3}}{2} \frac{(-1)^{n+1}}{\pi(n-\frac{1}{3})} \quad \text{or} \quad \text{Sinc}\left((n-\frac{1}{3})\pi\right)$$

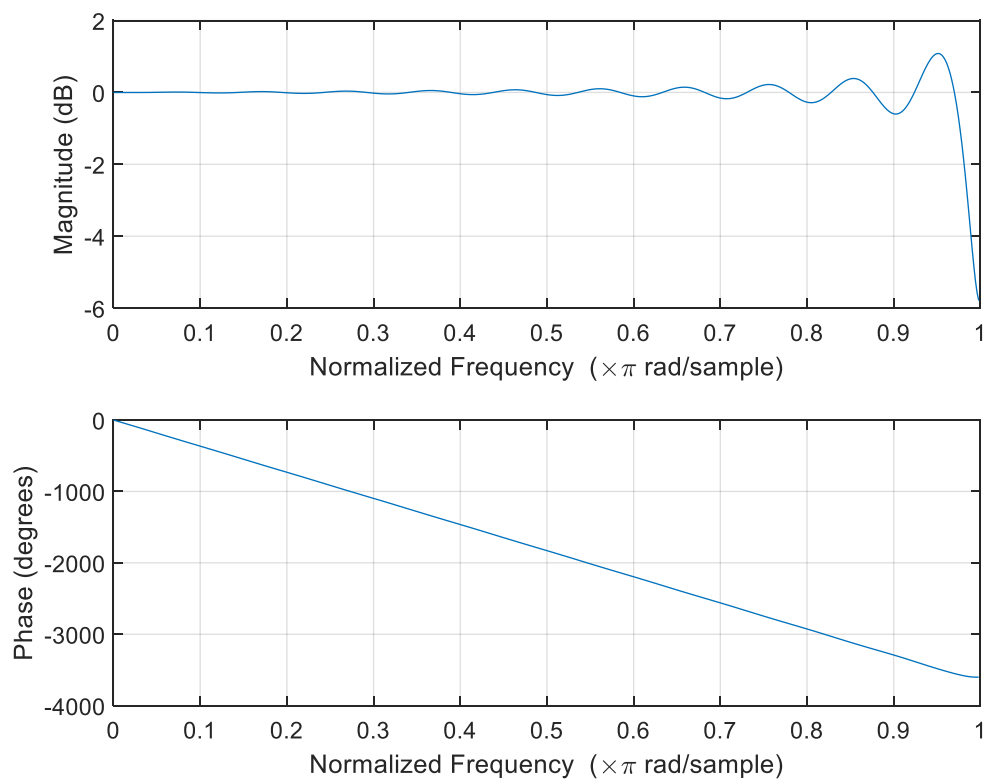
(b)



It resembles a sinc function.

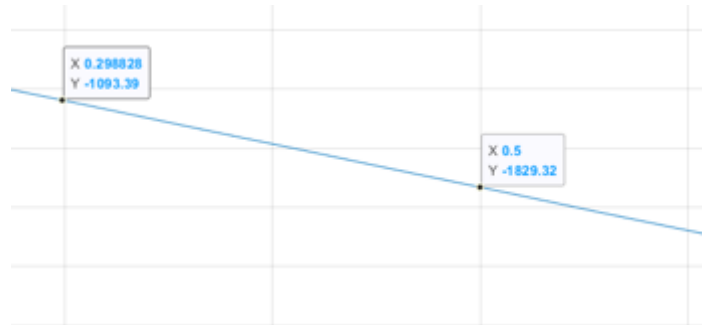
(c)

We know a sinc function applied Fourier transform is a rectangle function (ideal). However, now the signal is finite, in other words, the finite signal can be viewed as an infinite sinc function multiplied by a window in time domain. Hence, in frequency domain, it equivalently convolves with a sinc function, so we can see some ripples in the following picture and intensity of $\exp(-j\omega/3)$ is always equal to 1 or 0 dB. If we increase length of signal, e.g. $n = -100:100$, it's ripple will effectively mitigate and becomes more likely a rectangle function.



The slope of the phase spectrum can be derived from

$$Slope = \frac{phase \left(\frac{rad}{s} \right)}{\Delta frequency (rad)}$$



$$S = \frac{(-1829.32 - (-1093.39)) \times \frac{\pi}{180}}{(0.5 - 0.298828) \times \pi} = -20.3234$$

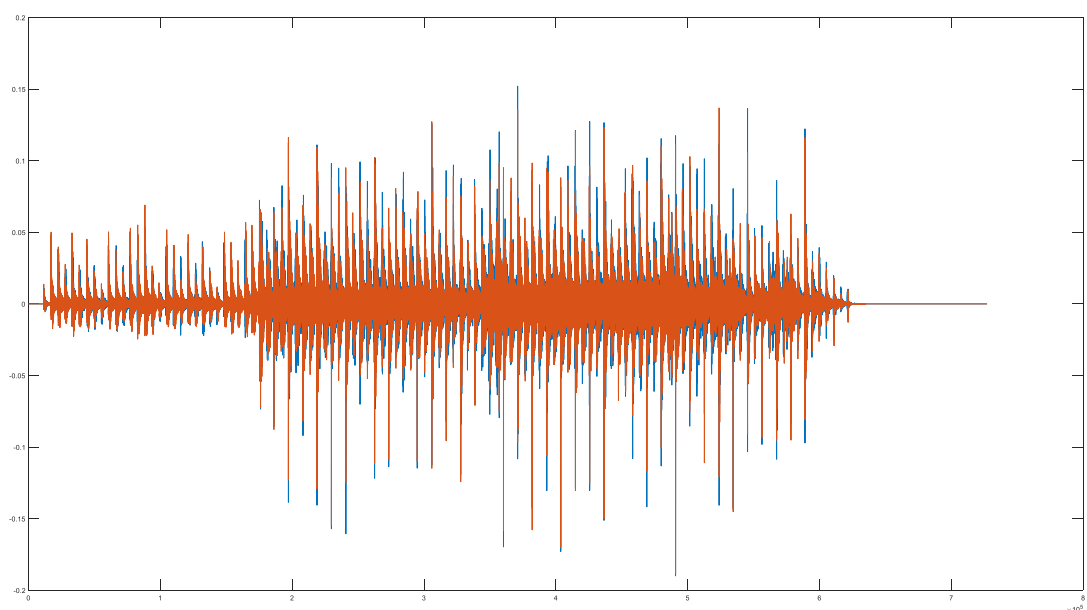
It's not as our expected value. The reason is the built-in function, `freqz`, implementing Fourier transform of $h[n]$ starts from 0 to 40 in time domain. In other words, it transforms $h[n-20]$ to the frequency domain rather than $h[n]$, so the phase is shifting by 30 rad/s.

4.

(a)

Compared to the original music, the first modified music is more terse and loses some tunes. Because we drop information once every two samples and retain its sampling rate. The second one and the third one are similar to the original tune. Due to the half sampling rate, its length is equal to the original one. I can't even hear their difference, the hearing may depend on something, e.g. earphone, sound card and so on.

(b)



Although their amplitude or intensity are slightly different, I think it can't be noticed easily due to our

degenerate hearing.

(c)

I can hear the difference for resampled music with half and one fourth sampling rate. The latter one sounds like muffled and seems to lack high frequency components. If sampling rate is not high enough, music would be distortion such that we can notice some peculiar tunes.